

In accordance with the invention, this object is attained by means of a novel yarn guide for the traversing feeding of a yarn to a rotatingly driven winding bobbin for producing a cheese in a winding head of a textile machine. In accordance with the present invention, the yarn guide is directly connected with at least one mechanical energy storage device which is permanently coupled with the yarn guide during the entire traversing movement for aiding the reversal of the direction of movement. The potential energy of the mechanical energy storage device continuously increases in the direction toward the dead points of the traversed area. As a result, the mechanical energy storage device together with the yarn guide constitute an essentially harmoniously oscillating mechanical system.

The invention is advantageously further developed by the below-described additional features.

The direct connection of the yarn guide with a mechanical energy storage device results in that no indirect transfer of forces is necessary at all. The only mass creating a moment of inertia is constituted by the yarn guide itself. By means of this the dynamics of the system are clearly improved over the prior art. The forces and moments of inertia in the area of the considerable angular accelerations at the reversing points are very small. It is possible to achieve a very rapid reversal of the direction of movement in the area of the end areas of the traversing movement without a considerably energy outlay.

In this connection it should be pointed out that in accordance with the present invention the entire oscillating body consisting of the yarn guide finger, the shaft supporting it, as well as the coil used as drive mechanism and its frame, are called the yarn guide.

It is possible to change the traversing width exclusively by means of the drive mechanism, i.e. by means of control or regulation. No mechanical adjustment at all is required for this if the energy storage device is designed for a maximum yarn guide stroke. Since with an increased stroke the torque required in the area of the reversal of the movement direction is also increased, the greater potential energy of the energy storage device being created in case of an increased stroke is useful. The noise level is lowered by the jolt-free operation of the yarn guide, and its service life is increased. Moreover, the regulating quality of the control is improved and the precise keeping of the traversing width is simplified.

The employment of a torsion spring, in particular a helical spring, is particularly advantageous in connection with the yarn guide of the invention. With an oscillation amplitude of, for example 60 to 70E, applicable here, such a helical spring also has the same progression of the characteristic curve of the spring to the left and right of the zero point. However, two helical springs can also be employed, which can then additionally be used for the electrical current supply for a coil of a drive mechanism of the yarn guide.

If the helical springs have opposite winding directions, the same spring torque progression is achieved in both oscillation directions in every case. However, as already mentioned, this advantage gains importance only if the angle of the deflection of the yarn guide out of the position of rest exceeds a threshold value, because in that case the progression of the characteristic curve of the spring is different during opening and closing of the spring.

While the energy storage device reduces the area under the graph of the drive torque square up to a third of the value which would be necessary without such an energy storage device anyway, setting the resonance frequency of the oscillating system to the upper limit of the required oscil-

lation frequency leads to the support of the oscillating motion, and therefore the relief of the electromechanical drive mechanism, being especially intense during extreme situations.

It is assured by the sturdiness of the arrestment of the energy storage device, that the energy storage device aids the oscillating motion in both directions to the same extent.

There is, of course, also the possibility to make the selection of the springs from the start in such a way, that their characteristic spring curves corresponds as much as possible to the desired oscillation behavior. In this connection it is even desirable that the characteristic spring curve does not extend in a straight line, but instead progressively increases after the reversing point, as already mentioned. The dimensioning of the spring should be selected for optimization in such a way that the integral under the driving torque square becomes the smallest. Because of this, the required power consumption of the drive mechanism for the yarn guide is the lowest.

The flexibility of the oscillation system can be improved still further if, in addition, the suspension of the at least one spring during the oscillation movement of the yarn guide is adjusted in a controlled manner. In a range of a lesser oscillation frequency, for example 5 Hz, the spring assistance in the reversing area of the yarn guide is required not at all or only to a small extent. The energy for bending the spring or, if an additional spring is used, the two springs, must be supplied by the electrical drive mechanism of the yarn guide in the remaining oscillation range outside of the oscillation dead center points, in which no angular acceleration is present. This energy can be reduced in that the suspension of the at least one spring for all practical purposes oscillates synchronously. This means that at very low oscillation frequencies the suspension could also oscillate with the same amplitude. However, in this case the drive mechanism for the suspension must be included when determining the energy balance. It can therefore be advantageous, depending on the actual conditions, to let the suspension oscillate along at the same frequency, but with reduced amplitude. In that case the bending of the spring takes place only in the lower range of the characteristic curve of the spring.

With increasing frequency, the requirement for assisting the movement reversal increases. For this reason a relief of the spring in the area of the reversing points is disadvantageous. The potential spring energy stored in the area of the reversing points can be clearly increased by appropriate phase-shifting between the oscillating movement of the suspension and of the yarn guide. However, in this way merely a far-reaching effect on a fixed frequency would be achieved, which could also be achieved with a stronger spring or a plurality of springs.

But the flexibility of the system offers the possibility of relieving the spring over the larger part of the stroke by an oscillation of the suspension in the same direction, and to clearly increase the spring force only directly prior to reaching the reversing point by swiftly reversing the a oscillating movement of the suspension into the opposite direction.

A "weak spring" is created in this way over the larger portion of the stroke length, and a very strong spring in the reversing points. This exactly corresponds to the desired assistance of the oscillating movement of the yarn guide. However, in the course of this, the system is not disadvantageously affected by an abrupt introduction of a spring force or adjustment of a spring constant, as in the prior art already